GH-18 House Model — Handover for the next AI agent or developer

Project: GH-18, Celebrity Homes, Palam Vihar, Gurugram, India. Renovation of an existing house, plus a proposed second floor. **Client / owner:** Mr. Praveen Kumar Yadav. The person giving instructions in the build sessions is the owner's side (the repository owner, `aditya-pandey`). **Repository:** `github.com/aditya-pandey/interior-design`. **Branches:**

- `main` is the repository's default branch and the live branch. GitHub Pages and Netlify both deploy from it.
- `house-design-v1` is the working branch. Changes are made there and merged into `main` to go live.
- All history up to this handover is on both. The first Claude session worked on an auto-named branch, `claude/zen-shannon-ilvm07`, which was then the default. It was renamed to `house-design-v1` and `main` was created from it.
- The old branch is identical to `main`. The owner is deleting it in the GitHub UI, because the agent's git connection could not delete it.
- An open Imgbot pull request (image compression) targeted the old branch and closes with it. It isn't needed.

Live sites:

- GitHub Pages: https://aditya-pandey.github.io/interior-design/
- Netlify, from the same repository's `main` branch (production branch set to `main`). Netlify sometimes lags behind GitHub Pages.

State at handover (25 Oct 2026): everything described here is committed and pushed. The last change is F16, the materials and quantities schedule with its 3D view, on `house-design-v1`; it is merged into `main` once the owner approves it. All automated checks pass, and GitHub Pages deployed successfully from `main`.

This document is meant to be pasted into a new Claude (or other agent) session as the opening context. It tries to carry everything the previous sessions learned: the goal, the data model, the code, the design decisions and why they were made, the user's preferences, the traps, and the open items. If anything here disagrees with the code, trust the code and fix this document.

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0. Read this first: the nine rules

1. **`house.json` and `house_r1.json` are the single sources of truth.** Plans, checks, the 3D viewer and the film are all generated from them. Never hand-edit the generated outputs (`plans/*.svg`, reports, `dist/`).
2. **Neither JSON is hand-edited either.**
   - `house.json` is written by `tools/build_house.py`.
   - `house_r1.json` is written by `tools/build_r1.py`, which deep-copies the main model and applies tagged patch functions.

   - To change the main site, edit `build_house.py`. To change Revision R1, add or edit a patch in `build_r1.py`. Then regenerate.
3. **Never change the main site (`/`) unless the user explicitly asks.** The owner's Oct 2026 changes live only in R1 (`/v1/`). After any tool or viewer change, check that the main outputs come out byte-identical; `git status` should show nothing changed under `plans/` (main).
4. **Units are mm in the model.**
   - R1 displays feet-inches with metres in brackets, e.g. `9'-1" (2.77 m)`. Use that style in anything the user reads about R1.
   - The main site displays metres.
5. **Orientation differs between the versions.**
   - Main site: the front (+y, street) faces EAST (`meta.north_vector [1,0]`; an early assumption).
   - R1: the front faces NORTH (`north_vector [0,1]`), confirmed by the user and by all consultant drawings. In R1, **east is −x** (the left, small-x side wall) and **west is +x**.
   - Vastu reasoning in R1 must use this frame.
6. **"No compromise on quality."** The user's words. Verify every visual change by rendering it (headless Chromium, see §9) and looking at the images before you push. Re-run every check script.
7. **Big changes are proposed first, then built step by step.**
   - The user often says "Do not implement anything. We will brainstorm first". Write a clear proposal, wait for "go ahead", then build in steps (A, B, C…), showing each one.
   - When a request is ambiguous or self-contradictory (for example "mandir near the lift in the north-east", when the lift is on the west), ask one crisp question with the options and the trade-offs.
8. **Work on `house-design-v1`; `main` is live.**
   - Commit to `house-design-v1` with descriptive messages.
   - When a change is verified and the user approves it, merge it into `main`. A push to `main` that touches `gh18-house/**` triggers the GitHub Pages deploy.
   - Do not open a PR unless asked. Never rewrite `main`'s history.
9. **Keep the records in step with the model.**
   - Every change gets a row in `ASSUMPTIONS.md` (sections A–E).
   - For R1, also update `CHANGE_REGISTER.md`, which maps every owner, consultant and Vastu item to what R1 does.

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1. What the project is

- An accurate, to-scale digital model of the house, built from the architect's PDF (`inputs/GH-18 GF FF Floor Plan.pdf`, drawing UPS/GH18/PV/001 R0, 17.08.25, "scale NTS").
- On top of it:
  1. **2D plans** (SVG) with dimensions, area tables and a dimension check against every string in the PDF.
  2. **A three.js 3D web viewer**:
     - dollhouse orbit, cutaway, exploded floors
     - first-person walk with collision, phone touch controls
     - day/dusk lighting, room labels
     - glTF props (the family's car)
     - Revision R1 with plumbing and electrical services in 3D
  3. **A 54-second cinematic film**, rendered headless from the viewer.
- Phase 3 (Blender renders) was planned in the original brief but never started.

Timeline of the work, so you know the history

1. **Phase 1:** data model, plans and dimension check (ASSUMPTIONS sections A and B).
2. **Phase 2:** the viewer (section C), then the film.
3. **Deployment:**
   - Netlify initially failed because the app is not at the repo root; `netlify.toml` fixed this.
   - GitHub Pages via an Actions workflow.
   - At the end, the branches were reorganised: `main` became the live default branch and `house-design-v1` the working branch (see the top of this document and §9).
4. **Client feedback round 1** (section D, main site):
   - D-01 closed kitchen
   - D-02 separate wash area
   - D-03 bedroom-3 toilet on the outer wall
   - D-04 kids room and bedroom 3 swapped (study desk faces east)
   - D-06 the family's car
5. **Site polish:**
   - site name, favicon set, Open Graph preview image
   - mobile UX: bottom-sheet menu and touch joystick
   - Baleno car model, recoloured grey and optimised
6. **Revision R1 (Oct 2026, section E):**
   - **Inputs:**
     - the owner's 12-point change list
     - three consultant mark-ups on R0: the Vastu red-line register, Electrical Layout Rev A and Plumbing Layout Rev A
   - **Process:** the user asked to brainstorm first, then approved a brief and asked for a separate URL (`/v1/`).
   - **Steps, each pushed and reported:**
     - A: orientation and units
     - B: ground floor
     - C: first floor
     - D: second floor
     - E: services (E12 plumbing, E13 electrical, E14 change register)
   - **Owner follow-ups:**
     - the mandir moved from the guest room to the family lounge (B6 revised)
     - a tulsi added (B15), first in the front lawn's NE, then moved to the back garden at the owner's request
7. **This handover document.**

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2. Repository layout

```
interior-design/                        (repo root)
  netlify.toml                          Netlify: base gh18-house/viewer, npm run build, publish dist, NODE
20, PUBLIC_BUILD=1
  .github/workflows/pages.yml           GitHub Pages: on push to main touching gh18-house/**
  gh18-house/
    house.json                          MAIN model (generated by tools/build_house.py)
    house_r1.json                       R1 model (generated by tools/build_r1.py)
    boq_r1.json                         R1 materials and quantities per room / category (generated by
tools/boq.py; read by the viewer)
    README.md                           how to run everything, conventions
    ASSUMPTIONS.md                      every guess, deviation and change: A (PDF reading), B (brief),
                                        C (viewer), D (client round 1), E (R1 table, incl. F16 materials)
    CHANGE_REGISTER.md                  R1: owner items, red-lines, plumbing / electrical clouds -> R1, holds
    HANDOVER.md / HANDOVER.pdf          this document
    inputs/                             architect PDF, 300 dpi raster, reference PNGs, pdf_dims.json
    plans/                              MAIN: GF.svg FF.svg SF.svg check_report.txt (generated)
    plans/r1/                           R1: GF.svg FF.svg SF.svg check_report.txt services_report.txt
(generated)
                                        + GH-18_R1_materials.xlsx / .pdf, materials.html (generated by
tools/boq.py)
    tools/
      houselib.py                       geometry helpers + ftin / feet / length_label / point_in_poly
      build_house.py                    writes house.json (W, ARC, O, F, CEIL helpers; importable, writes
only in __main__)
      build_r1.py                       writes house_r1.json = deepcopy(build_house.house) + PATCHES
      check_dims.py                     model vs every PDF dimension string (TOL 50 mm); [house_r1.json] arg
      check_services.py                 R1 services: PLUMBING, ELECTRICAL and VASTU PLACEMENTS rules +
schedules
      boq.py                            materials and quantities (tiles, pipes, wires, paint, POP, granite,
timber, MDF, lights)
      plans.py                          SVG plans; args: [house file] [out dir]
      pdf_extract.py                    re-reads dimension strings from the PDF (needs pdfplumber)
      gen_textures.py                   procedural CC0 PBR textures (used because Poly Haven was blocked)
      fetch_polyhaven.py                optional: swap in real Poly Haven textures/HDRIs where internet
allows
    assets/                             served as the viewer's public dir
      textures/ (+manifest.json)  models/baleno.glb (3.7 MB)
      site/ (favicon.svg, PNG icons, og.jpg, site.webmanifest)    LICENSES.md
    viewer/                             Vite 5.4.11 + three 0.170.0 (Node 20)
      index.html  vite.config.js  package.json
      src/main.js         app: renderer, UI, view logic, presets, walk, services UI, scripting API
      src/build.js        house JSON -> merged meshes per floor|category|material; lights; labels; presets;
colliders
      src/furniture.js    parametric furniture builders (B.<type>)
      src/materials.js    materials (textured + flat + emissive)
      src/geom.js         geometry helpers   src/walk.js  first-person walker   src/context.js  film-only
neighbourhood
      src/services.js     3D services layers (plumbing + electrical)
      src/style.css
      scripts/shots.mjs   screenshots of every preset (npm run shots)
```

```
  scripts/html2pdf.mjs  HTML -> PDF with Chromium (used for the materials PDF)
  film/              shots.mjs, render.mjs, post.py, score.py (cinematic film pipeline)
renders/                          film output location (the mp4 is not committed)
```

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3. How to run

```
# Python 3.9+, standard library only (pdfplumber only for pdf_extract.py)
cd gh18-house
python3 tools/build_house.py            # -> house.json           (main site model)
python3 tools/check_dims.py             # -> plans/check_report.txt (must say RESULT: PASS)
python3 tools/plans.py                  # -> plans/GF.svg FF.svg SF.svg

python3 tools/build_r1.py               # -> house_r1.json        (R1 = main + patches)
python3 tools/check_dims.py house_r1.json        # -> plans/r1/check_report.txt
python3 tools/plans.py house_r1.json plans/r1    # -> plans/r1/*.svg
python3 tools/check_services.py house_r1.json    # -> plans/r1/services_report.txt (exit 1 on any FAIL)
python3 tools/boq.py house_r1.json               # -> boq_r1.json, plans/r1/GH-18_R1_materials.xlsx, materials.html
(cd viewer && node scripts/html2pdf.mjs ../plans/r1/materials.html ../plans/r1/GH-18_R1_materials.pdf)

# Viewer (Node 18+, 20 used in CI)
cd viewer && npm install
npm run dev          # http://localhost:5173  (?rev=r1 for R1, ?low for weak GPUs)
npm run build        # dist/ with the main site at / and R1 at /v1/
npm run preview      # serve dist
npm run shots        # screenshots of every preset into viewer/screenshots/
PUBLIC_BUILD=1 npm run build   # what CI does: strips the client name/address from the JSON in the bundle
```

The R1 checks pass at handover:

- `check_dims`: `RESULT: PASS (0 mismatches over 50 mm, 0 unmapped PDF strings)`.
- `check_services`: `RESULT: PASS (278 pass, 4 warn, 0 fail)`. The four warnings are listed in §7.6.

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4. The data model (house.json / house_r1.json)

4.1 Frame and units

- **Units:** mm.
- **Origin:** the rear-left outer corner of the plot (outer face of the rear boundary × outer face of the left side wall).
- **Axes:**
  - **x:** across the plot, 0 → 9381 (`PLOT_W`). Side walls have centrelines at x 115 and x 9266 and are 230 thick, so the interior faces are x 230 and x 9151.
  - **y:** from the rear boundary toward the street. The rear glazing line is at y 5590; the front boundary's outer face is at y 22276 (`plot.depth_to_front_boundary`). The ramp lies outside the plot, beyond that.
  - **z:** up from natural ground (NGL = 0).

- **Levels (`levels`):** GF finished floor +450, FF +3750, SF +7050 (proposed), roof slab top +10350.
  - Floor to floor 3300; slab 300; clear height 3000.
  - False ceilings (POP) at 2750 above FFL, or 2600 in wet rooms.
- **Outdoor levels:**
  - lawns and parking: 0
  - deck, utility corner and entrance court: +300 (one step below the house)
  - parapets: 1050 high; boundary walls: 1500 high
- **Orientation:**
  - Main site: `meta.north_vector = [1,0]` (front faces East).
  - R1: `[0,1]` (front faces North, so East = −x and West = +x).
  - `meta.north_label` and `meta.display_units` (R1: `"ft-in (m)"`) drive the plans and the viewer.

4.2 Top-level keys

`meta, levels, schedule (doors/windows from the PDF), finish_schedule_pdf, materials, stair, slabs, floors[], services (R1 only)`

- `meta`: project, client, source, units, origin, axes, pdf_scale_mm_per_pt 28.53, north_vector, plot, revision (R1), display_units.
- `floors[]`: one entry each for GF, FF and SF:
  - `walls[]`, `rooms[]`, `outdoor[]`, `furniture[]`
  - optional `joinery[]`, `pergola` (GF), `canopy` (SF)

4.3 Walls and openings

- **A wall** is either `{id, a:[x,y], b:[x,y], t, kind, status, openings[], note}` or an arc `{arc:{c, r, from, to}}` (degrees), plus `t`.
  - `kind`: external | internal | lift | shaft | boundary | parapet | glass_wall | glass_partition | glass_railing | railing | screen.
  - `status`: existing (dark grey in the PDF) | new (light grey) | proposed. Plans draw these differently.
  - Optional `h` and `base` override the height.
- **An opening** sits inside its wall: `{id, mark, type, offset (mm along the wall from a), width, height, sill, hinge a|b, swing}`.
  - `swing` is the world direction the leaf opens into (`+x`, `-y`, …), or `out` for the pivot door.
  - Types: door, sliding_door, glass_door, pivot_door, window, lift_door, curtain, gate.
  - Moving a wall moves its openings.
- **Room polygons are explicit.** If you move a wall, update the polygons of the rooms it bounds. `check_dims` catches dimension drift.

4.4 Rooms and outdoor areas

- **Rooms:** `{id, name, kind?, polygon, floor finish, wall finish, ceiling:{type, height, cove}, status, note, label_at?}`.
  - `kind` may be shaft, lift or outdoor (SF terraces are rooms with kind outdoor).
  - IDs: `GF-KIT`, `GF-DIN`, `GF-DRW` (Family Lounge in R1), `GF-LOB`, `GF-TLT`, `GF-GST`, `GF-FOY`, `GF-MND` (Guest Lounge/Study in R1), `GF-JAN` (Under-stair Storage in R1), `GF-LFT`, `GF-OTS`, `FF-MST`, `FF-DRS`, `FF-MTL`, `FF-BAL`, `FF-TL3`, `FF-LOB`, `FF-BD3`, `FF-KID`, `FF-KTL`, …, `SF-LOB`, `SF-SWR`, `SF-WSH` (Store), `SF-SRV`, `SF-TRB`, `SF-TRF`.
  - R1 renames keep the old name in `was` and mark `brief_change: true`.
- **Outdoor areas** (`floors[].outdoor[]`): `{id, name, polygon, z?, finish}`.

- GF: `GF-LAWN-B` (back lawn, y 200–2814), `GF-DECK` (deck + pergola, y 2814–5515), `GF-LAWN-F`, `GF-COURT`, `GF-STEP`, `GF-PARK`, `GF-DRAIN`, `GF-RAMP`.

4.5 Furniture

- **Format:** `{id, room, type, pos:[x,y] (centre), size:[w,d,h], rot (deg), z? (raised above the floor), note, ...extras}`.
- **The piece's local +y is its FRONT.** So `rot` 0 faces +y (street), 90 faces −x, 180 faces −y (lawn), 270 faces +x.
  - Front vector in plan: `(−sin rot, cos rot)`.
  - The user (or devotee) stands in front and faces the opposite way.
- **Extras in use:**
  - `src: 'models/x.glb'` with `model_front: '-z'`: a glTF prop. The car `GF-007`, type `car_model`, has an empty builder.
  - `on_roof: true` with `z` = roof height above the SF: drawn in the ROOF layer, and dashed "ON ROOF" on the SF plan.
  - `chaise`, `sink: true` (utility counter), `credit`.
- **Furniture placed in an outdoor area** stands at that area's `z`.

4.6 Services (R1 only): `services.plumbing` and `services.electrical`

- **Common:** points are `[x, y, z]` with z ABSOLUTE in mm (GF FFL = 450).

Plumbing

- **Stacks:** `{id, kind, dia, at:[x,y], z:[z0,z1], serves, note}`.
  - SS-1 at (450, 11100) in the OTS, rising to 1 m above the roof.
  - SS-3 at (8950, 15420), in a boxed duct in the guest lounge's SW corner. Its vent sits above the front terrace parapet.
- **Chambers:** `{id, type: inspection|grease_trap|cleanout|rwh_pit|tank, at, size, depth, area, note}`.

Chamber	Position	Area	Notes
IC-2	(150, 11700)	OTS	
GT-1	(150, 12350)	OTS	
CO-1	(4800, 14200)	lobby	sealed cleanout
CO-2	(4800, 18000)	lobby / foyer	sealed cleanout
IC-3	(5300, 20300)	entrance court	
IC-4	(8700, 20300)	parking	heavy-duty cover
IC-5	(7600, 21900)	parking	heavy-duty cover; the only municipal connection
RWH	(4500, 21400)	entrance court	1200 dia pit
UGT	(900, 21100)	NE front lawn	1500 × 1200, 5000 L

- **Runs:** `{id, kind: soil|waste|sewer|cold|hot|pumped|rain, dia, pts[[x,y,z]], serves, outfall?}`.
  - Gravity runs are made by `drain()` in `build_r1.py`. It takes the start invert and computes every later z from the plan length × fall, rounding down per segment.
  - Falls are the minimums × 1.1: soil 1:40, waste 1:50, 150 sewer 1:50, 110 sewer 1:40, rain 1:100. The z you type after the first point is ignored.
- **Fixtures** come from the furniture, e.g. `wc_wall_hung`, `vanity`, `basin`, `sink`, `shower_head`, `bathtub`, `washer_dryer`, `utility_counter` with `sink`.

Electrical

- **points:** `{id 'GF-E01'…, kind, floor, room, serves, at, base, note, face?:[nx,ny], circuit, tag?, tr?, kw?, ceiling?}`.
  - `room` is where the point is. `serves` is the room it is for; switchboards outside wet rooms carry the wet room here.
  - `base` is the floor or area level.
  - `face` is the unit plan normal into the room, for wall-mounted symbols.
  - `tr` is the AC tonnage; `kw` is used for pumps and the lift.
- **kinds:**
  - lights: `light`, `pd` (pendant), `fan`, `ucl` (under-cabinet light), `ml` (mirror light), `wl` (wall light), `gl` (garden bollard)
  - switches: `sb`, `sb2w` (2-way / bedside)
  - sockets: `s6`, `s6c` (counter), `s16`, `usb`, `tv`, `data`, `ip65`, `ups`, `ch` (chimney)
  - big loads: `ac`, `gy` (geyser), `ef` (exhaust fan), `ev`, `iso` (lift isolator), `pump`, `spare`
  - low voltage: `cctv`, `vdp`, `vdpo`, `rack`
  - detectors: `sd`, `hd`
  - boards: `db`, `mtr`, `inv`
- **circuits:** `{id, cls: lighting|sockets|power|geyser|ac|ups|data, db, points[], mcb, cable, rcd, backup?, note?}`.
  - They are grouped automatically per DB: lighting ≤ 10 load points (switchboards don't count), sockets ≤ 8, and one dedicated circuit per 16 A outlet, AC, geyser, EV, pump and the lift.
  - DB-1 serves GF and outside, DB-2 FF, DB-3 SF and roof; LIFT is on DB-1. UPS and DATA run from the inverter / rack.
- **routes:** `{circuit, cls, trunk[[x,y,z]], drops[[[..],[..]]]}`.
  - Loop-in conduit runs nearest-neighbour from the DB, Manhattan legs in the slab above (GF circuits at 3600, FF at 6900, SF at 10200), with a vertical drop to each point.
  - Outdoor circuits run buried at −300.
- **submains:** SM-1 MTR→DB-1, SM-2 DB-1→DB-2, SM-3 DB-1→DB-3, SM-4 DB-1→INV, each hand-routed.
- **refrigerant:** `{ac, odu (furniture id), pts, length, note}`, hand-routed per AC.

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5. The tools in detail

5.1 `build_house.py` (main site)

- **Helpers** (also imported by `build_r1.py` as `base.F`, `base.O`, …):
  - `W(id,a,b,t,kind,status,…, openings)` for walls, `ARC(...)` for arcs
  - `O(id,mark,type,offset,width,height,sill,hinge,swing,…)` for openings
  - `F(id,room,type,pos,size,rot,z,**kw)` for furniture
  - `CEIL(type,height,cove)` for ceilings
- **Constants:** `XL_IN=230`, `XR_IN=9151`, `XL_C=115`, `XR_C=9266`, `PLOT_W=9381`, `Y_REAR_GLASS=5590`, plus the arc centres fitted to the PDF vectors (max deviation 3 mm).
- **Writing:** the script writes only under `if __name__ == '__main__'`, so `build_r1.py` can import `base.house` without side effects. `dump()` gives a compact, stable JSON layout; keep it so diffs stay readable.

5.2 `build_r1.py` (Revision R1)

- **Process:** `h = deepcopy(base.house)`, set `meta.revision='R1'`, then run every function in `PATCHES` in order.

- **Helpers:** `floor(h,'GF')`, `find(items,id)`, `drop(items,*ids)` (asserts the ids exist), `opening(wall,id)`, `rename(room,name,was,note)`.
- **PATCHES at handover, in order:**

Patch	Item	What it does
`a1_orientation`	A1	north_vector [0,1], labels
`a2_units`	A2	display_units ft-in (m)
`b3_family_lounge`	B3	L-sofa faces the TV on the W wall; dining basin GF-F24
`b4_kitchen_wash`	B4	removes the D-02 wash yard; GL-4 back to R0; wash nook GF-K10/K07/K13 inside the kitchen; KW-1 window + exhaust into the OTS
`b5_guest_lounge`	B5	GF-MND → Guest Lounge / Study (desk on the E wall, sofa, TV, settee, coffee table, rug)
`b6_mandir`	B6 revised	open mandir GF-F25 on the lounge E wall at (3525, 8350), rot 270; pouf F26, rug F27; crockery F19 moves to (3525, 7000)
`b7_under_stair`	B7	storage units, inverter / network rack, door swing; DB cabinet K12 beside the fridge tower K04 on the kitchen E wall
`c8_pantry`	C8	FF lobby pantry counter FF-L01 + wall cupboards FF-L05
`c9_kids_toilet_back`	C9	reverts D-03: kids toilet KTL back beside the stair at its R0 size
`c10_master_suite`	C10	washroom and dresser swapped; tub FF-T01, WC FF-T02 by the OTS, MW-1 window; doors D2a / D3a re-hung
`d11_utility_terrace`	D11	SF utility on the covered back terrace; store; drying racks; ODUs; OHT SF-R01 and solar heater SF-R02 on the roof
`b15_tulsi`	B15	tulsi platform GF-O09 at (645, 1650), rot 270, in the back lawn's NE corner; bamboo planters O10 / O11; fluted panel O12 on the E boundary wall
`e12_plumbing`	E12	`services.plumbing`
`e13_electrical`	E13	`services.electrical`; also re-places the ODUs (drops SF-AC2, moves SF-AC1, adds SF-AC5/6 and GF-O08)
`f16_materials`	F16	kitchen, wash-nook and pantry counter tops get `top: 'granite_black'` (the BOQ's granite); quantities come from `tools/boq.py`

- **To add an R1 change:**
  1. Write a function with a docstring that states the item ID and the reasoning.
  2. Append it to `PATCHES`; order matters, so services go last because they read furniture positions.
  3. Rebuild and run all the checks.
  4. Regenerate the R1 plans.
  5. Render and look at the result.
  6. Add an ASSUMPTIONS section E row and a CHANGE_REGISTER row.
  7. Commit and push.

5.3 `check_dims.py`

- **What it does:** maps every one of the 51 PDF dimension strings (`inputs/pdf_dims.json`) to a probe on the model and compares within TOL = 50 mm. It also checks openings against the schedule, plus the brief's checks.
- **`SUPERSEDED`:** PDF strings made obsolete by a client change (D-01, D-03). They are listed as SUPER, not counted as failures.
- **`REVERTED = {"R1": ("D-03",)}`:** R1 undid D-03, so those four strings are measured again in R1, and they pass.

- **Exit code:** 1 on any mismatch or unmapped string.

5.4 `plans.py`

- **Arguments:** `[house file] [out dir]`. It sets the globals FEET and SRC.
- **What it draws:**
  - walls coloured by status (existing dark, new grey, proposed hatched)
  - openings with swings, opening tags, room labels with size and area, dimensions, outdoor areas
  - furniture symbols (`furniture_symbol()`; add a branch there for new types), the stair
  - overlays: floor above, roof items dashed
  - a title block with north arrow, scale bar, legend and room-area table
- **Units:** R1 labels are in feet-inches with metres in brackets.

5.5 `boq.py` (materials and quantities, F16)

- **Run:** `python3 tools/boq.py house_r1.json`, after `build_r1.py`. It writes `boq_r1.json` (used by the viewer), the Excel file and `materials.html`; `viewer/scripts/html2pdf.mjs` turns the HTML into the PDF. Commit all of them, because the viewer imports `boq_r1.json` and the Vite build copies the PDF and Excel into `dist/files/`.
- **The owner's nine items, measured from the model:**
  - **Tiles:** wet-room floors and walls (to the false ceiling, minus openings), deck tiles, the SF vitrified floor; boxes per tile size with wastage; waterproofing for wet rooms.
  - **Living floors, both ways:** A = Italian marble (+15%), B = 1600 × 800 vitrified (+8%), plus skirting. The owner decides later.
  - **Pipes:** lengths per kind and diameter along the plumbing runs and stacks (3 m lengths +10%, bends counted); chambers; AC copper pairs per route.
  - **Wires:** each circuit's route × 3 cores +10% (90 m coils), conduit; sub-mains (sizes indicative); Cat-6; accessory counts; DB ways.
  - **Paint:** interior walls (minus openings, tiles, cladding, fluted walls, backsplash), ceilings on POP, exterior (non-party walls, parapets, boundary walls); putty, primer and emulsion litres at indicative coverage.
  - **POP:** ceiling areas, gypsum boards, perimeter channel and cove lengths.
  - **Granite:** counter tops with cut-outs, window sills, door thresholds, the stair and flamed paving; the kitchen backsplash is listed as white marble (as designed).
  - **Timber:** door frames and shutters, pergola cft, handrail, veneer cladding (lobby, foyer, guest lounge), fluted walnut, the tulsi panel and the mandir.
  - **MDF:** per wardrobe and storage unit (carcass, shutters, back, sheets, hinges, edge band, laminate). BWP ply for wet units (vanities, sink base, utility, terrace), as the owner asked: MDF elsewhere.
  - **Lights:** downlights on the 1.4 m grid the 3D shows (each electrical "light point" switches a group), cove and under-cabinet LED, fans, mirror, wall and garden lights, exhaust fans and detectors.
- **Specs and rates live at the top of the file** (`TILE`, `OPTION_B`, `MARBLE`, `PIPE_SPEC`, `PAINT`, `CABLE_SUBMAIN`). Change a spec there and re-run. No costs are included (owner's choice).

5.6 `check_services.py`

It runs three sections and writes `plans/r1/services_report.txt`.

- **PLUMBING:**
  - every wet fixture is drained (WCs by soil pipes) and has a cold supply within 2.2 m
  - every drain ends at a stack, a chamber, another drain or a marked outfall
  - every gravity run falls at least at its minimum (`MIN_FALL` per kind and diameter)
  - chambers are not inside rooms (sealed cleanouts excepted)
  - nothing runs under the guest room

  - nothing is in the party walls (x 0–230 and 9151–9381, outside the OTS)
  - the UGT is ≥ 2 m from the sewer and the RWH
  - OHT head over the showers (warn under 2.5 m)
  - vents ≥ 1 m above the roof or terrace
- **ELECTRICAL:**
  - the sheet's point schedule, row by row, mapped to R1 rooms (`SHEET` table)
  - every room lit and switched, with its switchboard ≤ 1.8 m from a door (open-plan rooms and lobbies are switched 2-way and exempt)
  - wet rooms: switches outside; exhaust on the OTS or a non-party external wall, or a ceiling fan ducted to an open terrace; geyser ≥ 1.8 m high and ≥ 0.6 m from the shower or tub
  - sockets ≥ 0.6 m from basins, sinks, tubs and showers in the same room
  - every bedside table has a 2-way switch and a socket
  - ACs: not over glazing, ≥ 2 m high, refrigerant route to an ODU ≤ 15 m (warn up to 25 m), route clear of the party walls, every ODU used once
  - ODU Vastu zone (warn N / NE / E)
  - DBs: not in the Brahmasthan (centre ninth of the house footprint) or a wet room; SE preferred (warn otherwise); ≥ 0.6 m from water pipes
  - detectors: smoke in bedrooms, lounges and lobbies; heat, not smoke, in the kitchen
  - circuit limits and RCDs; every point on a circuit; circuits fed from their own floor's DB
  - no EF or ODU on the party walls
  - load: connected kW, demand at 0.55 diversity, so 3-phase
  - **Also prints:** the point schedule per room, the circuit table (MCB, cable, RCD, kW) and the AC routes. This part is meant for the electrician.
- **VASTU PLACEMENTS:**
  - mandir: the devotee faces E or N; it sits in the NE quarter of its room; no toilet behind, beside or above it
  - tulsi: in the NE corner of its garden; the devotee faces E or N; plot zone N / NE / E preferred (warn otherwise); open to the sky
  - UGT in the NE; OHT in the S / SW / W; the cook faces east
- **Zones:** Vastu zones use a 3 × 3 grid over the house footprint for the DB, ODU and OHT checks, and over the plot for the UGT and tulsi checks.

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6. The 3D viewer in detail

6.1 Pages and flags (`vite.config.js`)

- **One build serves the main site (`/`) and R1 (`/v1/`).**
- **The `extraPages` plugin** copies `dist/index.html` into each subfolder and adds three things:
  1. a script that redirects to the trailing-slash URL
  2. `<base href="../">` (or `"../../"`), so assets resolve from the root and nothing is duplicated
  3. a `window.GH18_REV` flag
- **`main.js` picks the version from:**
  - `REV`: `window.GH18_REV`, or `?rev=r1`, or a `/v1/` path
- **Why `<base>` and not copying assets:** an earlier version 404'd on assets (the preload scanner resolved relative paths against the subfolder). The static `<base>` plus slash redirect fixed it.
- **`stripPrivate`:** with PUBLIC_BUILD=1 it removes `meta.client` and `meta.source` and sets `meta.project='GH-18'` in both JSON files.
- **`siteUrl`:** fills `__SITE_URL__` in the Open Graph tags (from SITE_URL, or Netlify's URL).

6.2 `main.js` flow

1. **Setup:**
   - Imports both JSON files and picks `house` by REV.
   - `FEET` comes from `display_units`. `TOUCH` = coarse pointer; `LOW` = `?low` or touch; `PHONE` = ≤ 640 px.
   - `CINEMA` (`?cinema`): no UI, bloom, camera driven by `viewer.cine()`.
2. **Renderer:**
   - WebGLRenderer with ACES tone mapping and a CSS2DRenderer for labels.
   - Sky, PMREM RoomEnvironment (neutral studio IBL), a directional sun with shadows, a hemisphere light.
   - `NORTH` / `EAST` come from `meta.north_vector`. `sunDir(az, el)` uses them, so the sun is correct in both versions. `compass()` names the exterior views.
3. **`init()`:**
   - `createMaterials`, then `buildHouse(house, M)` returns meshes tagged `userData {floor, cat, mat}`.
   - One `THREE.Group` per FLOOR_ID (GF, FF, SF, ROOF); SITE meshes go straight in the scene.
   - Point lights per room (dusk only), room labels (CSS2D, feet-inches for R1), the Walker, `setupServices()` if `house.services` exists, `setupMaterials()` on R1, post-processing (GTAO unless LOW), `buildUI()`.
   - Portrait screens get a framed opening view. Then it sets `window.viewer.ready = true`.
4. **`applyView()`** applies `state {floor, explode, cutaway, mode, time, labels, services, materials}`:
   - floor visibility and explode offsets (4.5 m per floor)
   - ceilings shown only in walk mode
   - cutaway ghosts the `upper` category (walls above 1.2 m)
   - labels
   - services ghost every building mesh: `ghostOf(material, opacity)`, keyed by material uuid + opacity; opacities in `SERVICE_GHOST`: site 0.4, floors and slab caps 0.16, furniture 0.16, others 0.1
   - only the active services layer is shown; a picked floor isolates that floor's services (ROOF goes with SF)
   - the legend HTML is swapped per layer (services or materials)
   - **one pass sets every building mesh's material:** the original, or its Materials-view colour (`overlayOf`, by the mesh's material key); then ghosted for the services view (whole building) or the cutaway (`upper` walls). Keep this single pass when adding modes.
   - **Materials view (R1):** surfaces coloured by category (`MAT_CATS`: marble, tiles, deck, paint, granite, white marble, timber, MDF, POP); a quantity chip per room from `boq_r1.json` (`isMatLabel`, shown for the picked floor, decluttered like the service tags); the legend shows totals and links `files/GH-18_R1_materials.pdf` / `.xlsx`. Materials and services are mutually exclusive.
5. **Presets:**
   - Overview: Dollhouse, Exploded, GF / FF / SF cutaways, front and rear exteriors, plus for R1 "Services — plumbing", "Services — electrical", "Materials — ground floor" and "Materials — first floor".
   - One preset per room or outdoor area.
   - `framed()` pulls the camera back on portrait screens.
6. **Walk mode:**
   - Walker: PointerLock on desktop. On touch, the left 45 % of the screen is a joystick and the rest drags to look.
   - Eye height 1.6 m, collision against walls, glass and furniture, follows floors and stairs.
7. **UI** (`index.html` panel):
   - controls: View select, Floors GF/FF/SF/All, Explode, Cutaway, Room names, Materials & quantities (R1), Services Off/Plumbing/Electrical (only on pages with services), Light Day/Dusk, Walk
   - hints and the Baleno credit
   - **Keys:** 0–3 floors, C cutaway, L labels, P plumbing, E electrical (not in walk mode), M materials (R1), N day/dusk, Q leave walk.

- **Phones:** the panel is a bottom sheet with a Menu button; the walk HUD moves to the top; the services legend is a compact 3-column grid at the top.

8. `window.viewer` **scripting API** (used by every render or test script):

```
ready, headless (?headless: no animation loop; call settle()),
presets(), goPreset(id, instant), setFloor('GF'|'FF'|'SF'|'ALL'), setCutaway(b), setExplode(0..1),
setLabels(b),
setServices(null|'plumbing'|'electrical'|true), setMaterials(bool), setTime('day'|'dusk'),
walkAt(roomId, yawDeg, pitchDeg), walkPose(x_mm, y_mm, foot_m, yawDeg, pitchDeg), exitWalk(),
settle(n) (render n frames synchronously), stats(), cine({pos,target,fov,floor,cutaway,explode,time
0..1,exposure,interior,context,roll}),
renderFrame(), modelsReady (Promise for glTF props), error (set if init failed)
```

   - `walkPose` takes the foot height in METRES (0.45 = GF floor, 0 = lawn). Yaw: 0 looks −y, 90 looks −x (east in R1), 180 looks +y, 270 looks +x.

6.3 `build.js`

- **Merging:** batches geometry per `floor|category|material` and merges it with `mergeGeometries` for performance.
- **Categories:**
  - `struct`: walls below 1.2 m
  - `upper`: above 1.2 m; ghosted by the cutaway
  - `glass`, `glassUpper`
  - `ceiling`, `floor`, `slabcap`
  - `furn`
  - `emissive`: cove, lamp, jaali_glow and downlight materials; no shadows
  - `site`: ground, street
- **Builds:** walls with openings, frames, glass, doors (hinged doors shown open about 90°, sliders closed but walkable), room floors and ceilings (POP + cove + downlights), tile skins in wet rooms, stair, slabs, pergola, canopy, outdoor areas, furniture (`buildFurniture`), backsplashes, and the lights list (rooms; the pendant; `mandir_wall` / `mandir_open` / `tulsi_platform` glow).
- **Returns:** `{meshes, labels, presets, lights, models, colliders, …}`.

6.4 `furniture.js` conventions

- Every builder `B.<type> = (parts, w, d, h) => {...}` works in metres in the local frame:
  - X = width
  - Z = depth, with the FRONT at +Z
  - Y = up from the base
- **Helpers:** `box(parts, mat, w, h, d, x, y0, z, bevel)` (bottom at y0), `cyl(parts, mat, rTop, rBot, h, x, y0, z, seg)`, `sphere(parts, mat, r, x, y, z, scaleY, detail)`, `plateZ(...)` (a thin plate facing +Z), `legs4`, `grooves`, `cabinet`, `beds`, `sofaStraight`, `armchair`, …
- **Raw geometry:** push `{g, m}` with any THREE geometry.
- **Material keys:**
  - textured: marble_beige, marble_white, walnut, walnut_fluted, plaster, plaster_ext, tile_matte_lg, tile_matte_dark, vitrified, deck_tile, granite_flamed, cobble, grass, pebbles, concrete
  - flat: boundary, pop_white, slab_edge, black_metal, brass, steel, ceramic, linen, linen_white, fabric_taupe, fabric_olive, fabric_rust, fabric_blue, leather, rug, rug_dark, lacquer_white, lacquer_greige, terracotta, foliage, foliage_dark, mattress, appliance, car_paint, tyre, asphalt, ground, lift_steel, mirror, trunk

  - special / emissive: glass, glass_frosted, polycarbonate, cove, downlight, lamp, jaali_glow, window_lit, window_dark

- **To add a new furniture type:**
  1. Add a builder in `furniture.js`.
  2. Add a plan symbol branch in `plans.py` `furniture_symbol()` (otherwise it draws as a plain rectangle).
  3. If it glows at dusk, add a light in `build.js`.
  4. Render it in day and dusk.

- **Notable builders added in R1:**
  - `sofa`, `settee`, `mandir_wall` (withdrawn), `mandir_open`
  - `bathtub`, `water_tank`, `solar_heater`, `ac_odu`
  - `tulsi_platform`, `bamboo_planter`, `fluted_panel` (with a brass Om made of torus arcs)
  - `car_model` (empty: the glTF is loaded separately)

6.5 `services.js` (3D services)

- **`layer()`:** returns `add(fid, key, geometry)` and `finish(look)`. `finish` merges geometry per floor band and colour key into meshes.
- **Drawing order:** service materials are `transparent: true` with renderOrder 10 (9 for translucent chambers), so they draw AFTER the ghosted building. The pipes stay crisp, like an X-ray.
- **Floor bands:**
  - **Plumbing:** `BANDS` start 400 mm under each FFL (3350 / 6650 / 9950), so a floor's sunken-slab drains belong to that floor.
  - **Electrical:** uses `EBANDS` at the FFLs (3750 / 7050 / 10350), because a floor's conduits run in the slab above it. Single-floor circuits are forced onto their own floor; sub-mains, refrigerant and UPS/data runs split by height.

- **`buildPlumbing(pl)`:**
  - tubes per run, with joints
  - stacks with vent cowls
  - chambers as boxes, the RWH pit as a cylinder, with lids
  - tags for stacks and chambers
  - legend

- **`buildElectrical(el, names)`:**
  - symbols per kind (`symbol()`): ceiling discs, a 3-blade fan, pendant spheres, wall plates oriented by `face`, AC / geyser / DB boxes, garden bollards drawn from `base`
  - conduit polylines coloured by circuit class; sub-mains black; refrigerant cyan
  - tags for DBs, MTR, INV and each ODU (with the room it serves)
  - categories in `ECAT`

- **Tags:** `main.js` `declutterTags()` hides any tag that overlaps an earlier, higher-priority one, every frame.
- **Bug fixed, remember it:** `g.visible` must be a real boolean. `state.services` is null when off, and `null && x` is null. three.js only hides objects whose `visible === false`, so layers stayed visible until the flag was coerced with `!!`.

6.6 glTF car

- **Source:** `assets/models/baleno.glb`: the 2022 Maruti Suzuki Baleno by BHP3D, CC BY 4.0, recoloured to Grandeur Grey metallic.
- **Optimisation:** 708k → 262k triangles, meshopt compression, 31 MB → 3.7 MB.
- **Loading:** `loadModels()` uses GLTFLoader with MeshoptDecoder. The model is kept at native width and height and stretched along its length to `size[1]` = 3990 mm. With `model_front '-z'` it is turned 180°.
- **Credit:** must stay visible in the panel and in `assets/LICENSES.md` (licence requirement).

- **In `?cinema`:** the car meshes are hidden.

6.7 Film

- **Shot list:** `viewer/film/shots.mjs` holds camera paths in model coordinates, cut on the bars of a 96 BPM score.
- **Rendering:** `render.mjs` renders the frames headless (about 5 s per frame; resumable).
- **Post:** `post.py` does the edit, grade, letterbox titles, score (`score.py` synthesises it) and the encode.
- **Context:** `?cinema` adds a neighbourhood (`context.js`); it is scenery only.
- **Output:** 54 s, 1920 × 1080 at 24 fps, 2.39:1. The mp4 is not committed.

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7. Revision R1 — the design content and the reasoning

7.1 The owner's original change list, verbatim in meaning (Oct 2026)

1. Family lounge, TV, dining and a small washbasin.
2. Dimensions in feet instead of metres.
3. Servant-room washroom. The owner later said to ignore this one.
4. FF: instead of the cupboard in the lobby near the lift, a pantry slab with a storage cupboard.
5. Revert the bathroom layout changes of bedroom 3 / kids room.
6. A bath tub in the master bedroom's washroom.
7. Kitchen washing area: outer construction is not allowed.
8. Convert Mandir + Store into a work table, a sitting sofa with one settee (sethi), and a TV for guests.
9. Design a small mandir somewhere ("chotu sa kahin": a tiny one).
10. Plan storage beneath the stairs.
11. Master bedroom: exchange the dresser and the washroom area, with the door on the dresser side.
12. SF washing area: washing machine, ironing and utility, with a place to hang clothes on the terrace.

Plus the consultant drawings: wiring, plumbing and electrical.

7.2 Clarifications the owner gave (approved brief)

1. North = the road side.
2. Swap the master dresser and washroom, and add a tub.
3. No construction outside.
4. A small mandir per Vastu on the GF. This was **later revised**: not in a bedroom; in the family lounge's NE corner.
5. A pantry slab with storage above, near the lift.
6. Ignore the servant washroom; the SF utility should be "outside type" (washing machine, ironing).
7. Feet, with metres in brackets.
8. Show the services in the 3D view.
9. Keep R1 on a different URL (`/v1`) and don't touch the existing site.
10. **Later follow-ups:**
    - the mandir moved to the family lounge, east wall, NE corner, with the devotee facing east, after an Instagram reference photo
    - a tulsi was added after a reference photo: first in the NE front lawn, then moved by the owner to the **back garden**

7.3 Where things are in R1, and why

- **Family Lounge (GF-DRW + GF-DIN, open plan):**
    - TV and 2 TR AC on the west (fluted walnut) wall; the L-sofa faces it

  - pendant over the 6-seat dining table
  - hand-wash basin with a mirror light in the niche by the kitchen wall (it drains to the GF toilet line)
  - open mandir on the east (kitchen) wall at the NE corner, devotee facing east; the crockery unit is south of it on the same wall
- **Kitchen (GF-KIT):**
  - closed, with the KD-1 sliding door
  - hob on the east wall (the cook faces east, red-line 7)
  - wash nook on the north side with a double sink and an exhaust into the OTS (B4); prep counter at the south, under GL-4
  - fridge tower, then DB-1 in a 400 mm cabinet, on the east wall (SE zone)
  - the earlier D-02 outdoor wash yard is removed (owner item 7)
- **Guest Lounge / Study (GF-MND):** desk on the east wall (user faces east), 2-seater sofa facing a TV on the west wall, settee on the south wall. The SS-3 boxed duct is in its SW corner.
- **Guest room (GF-GST):** wardrobe on the south wall, bed headboard on the east wall, no mandir. Its AC sits over GL-1; the outdoor unit is on the front lawn (sheet A6).
- **Under-stair (GF-JAN):** storage cut to the soffit, plus a ventilated inverter / battery / network rack.
- **FF master suite:**
  - bedroom: king bed, headboard on the west wall
  - dresser in the old washroom zone: mirror on the east wall, DB-2 on its west wall
  - washroom in the old dresser zone: tub with an over-tub shower screen, WC on the east wall by the OTS on stack SS-1, vanity on the south wall
  - **Hold point:** this washroom sits over the kitchen, against red-line 11. It needs a new sunken floor and waterproofing.
- **FF lobby:** 1.5 m stone pantry counter with wall cupboards, 250 mm clear of the lift door (C8).
- **FF kids room and toilet:**
  - the kids toilet is back beside the stair at its R0 size, on SS-3 (C9)
  - the study desk faces north along GL-9
  - the TV sits in a niche in the wardrobe run
- **FF bedroom 3** (the front-left room in plan) has its own toilet on SS-1.
- **SF:**
  - servant room, servant washroom (unchanged)
  - store (the old washing room)
  - open utility on the covered back terrace: washer-dryer, utility sink, ironing, tall cupboard, drying racks
  - front terrace kept open
  - OHT 2000 L and solar heater on the SF roof
- **AC outdoor units:**
  - back terrace: SF-AC1 at the SE corner (bedroom 3, via the OTS, 14.3 m); SF-AC3 (family lounge, up the balcony west wall); SF-AC4 (master, straight up)
  - front terrace, NW against the stair wall: SF-AC5 (kids room, straight up); SF-AC6 (guest lounge, up the SS-3 duct)
  - front lawn: GF-O08 (guest room)
- **Tulsi:** back garden NE corner, against the east boundary wall, devotee facing east. Plot zone SE, flagged for the Vastu consultant.

7.4 Plumbing (E12) — reasoning

- **Drainage:**

    - Because of C10 the master WC sits by the OTS, so SS-2 is dropped and everything wet on the east side goes to SS-1 in the OTS.
    - The kitchen wash nook drains to grease trap GT-1 in the OTS, so the rear lawn has no chamber or pit (red-line 5, P1, P6).
- **Sewer route** (as in Rev A): IC-2 (OTS) → sleeved under the GF toilet, lobby and foyer with sealed cleanouts CO-1 / CO-2 → IC-3 (entrance court) → IC-5 (parking). SS-3 joins through IC-4. Only IC-5 connects to the municipal sewer.
- **Water:**
    - UGT in the NE front lawn, 3.4 m from the sewer and 2.2 m from the RWH pit.
    - The meter inlet does not cross the sewer.
    - Pumped riser R-1 runs in the east-wall chase above FFL (so it is not "under" the guest room), then up the OTS to the OHT on the SF roof.
    - DT-1 comes down the OTS; DT-2 goes to the SS-3 duct.
    - Solar hot water goes to the master washroom and the kitchen.
- **Rain:** four 100 mm RWPs on the front face to the RWH pit; its overflow goes to the 300 mm storm drain, which takes no sewage.
- **Warning:** the OHT base is only 1.45 m above the SF shower head, so the SF washroom gets a small pressure pump (the electrical sheet has a terrace pump point).

7.5 Electrical (E13) — reasoning

- **DBs:**
    - DB-1 (main) at (250, 9451), on the kitchen east wall, SE.
    - DB-2 at (2973, 5950), on the dresser's west wall. It is still in the SE zone, and it could not stay over DB-1 because the C10 washroom is there now.
    - DB-3 at (250, 13300), in the SF servant room for the SF and terrace. **Warn:** E zone.
    - Inverter + rack under the stair; meter + main switch on the front boundary wall at (5000, 22290), read from the street.
- **ODUs:** both side walls are party walls (with GH-17 and GH-19), so no ODU, exhaust or refrigerant penetration may be on them. That forced every ODU to the terraces or lawns, as listed in 7.3.
- **Changes from the sheet** (all recorded in ASSUMPTIONS E13 and CHANGE_REGISTER §4):
    - Bedroom-3 toilet switches are in the bedroom niche, because its door opens from bedroom 3, not the lobby.
    - Master washroom switches are at the dresser door, because wardrobes flank the washroom door.
    - Bedside points are at 600 mm, because the side tables are 500 mm high.
    - The guest AC is above GL-1 at 2450.
    - The kids-toilet EF is a ceiling fan ducted to the open terrace (its W wall is a party wall).
- **Load:** 45.9 kW connected, about 25 kW demand, so a 3-phase supply.

7.6 Materials (F16) — reasoning

- **What the owner asked:** "additions, need to get all of these in place": tiles, pipes, wires, paint, POP, granite, timber, MDF board in wardrobes, lights. Read as a materials-and-quantities schedule, and confirmed with the owner:
    - quantities only (no costs)
    - living floors both ways (marble or tiles) to decide
    - MDF as asked
    - a Materials view in 3D
- **Why measured, not typed:** every number comes from the model, so a design change and a re-run of `boq.py` keep the schedule right.

- **Downlights:** the electrical sheet's "light points" are switched groups. The fixture count follows the 1.4 m downlight grid that the 3D ceilings already show, so the 3D and the schedule agree.
- **Granite:** the counter tops were white marble in the model, but the owner listed granite (the norm for Indian kitchens), so R1 now shows 20 mm black granite and the schedule matches. Vanity tops and the backsplash stay white marble.
- **MDF in wet places:** plain MDF swells with damp. The owner chose MDF, so BWP ply is kept only for genuinely wet units. The note is in the Excel file.

7.7 Open items and hold points (also in CHANGE_REGISTER §5)

The four warnings in the services report:

1. SF shower head pressure → pump.
2. Guest-room ODU in the NE zone (the sheet's own A6).
3. DB-3 in the E zone.
4. Tulsi in the plot's SE zone (the owner's choice).

Hold points and open questions:

- **Master washroom over the kitchen:** sunken floor, waterproofing, flood test (architect, structural, plumber).
- **Living-area floors:** the owner still has to choose A (Italian marble) or B (large-format tiles); both are in the materials schedule.
- **Master headboard on the W wall** against red-line 13 (S wall). The electrical sheet assumes W.
- **SF rooms not in the S / SW** (red-line T); the SF is a proposal.
- **Structural:** the deck pergola kept or removed under "no outside construction" (asked, unanswered); the tank stand; the OTS opening; the lift pit and overrun; the balconies; the front curved balcony (red-line 16, P9), which is not in the model.
- **Plumber:** sewer invert levels and the municipal connection (RWA / MCG); the GF toilet vanity at the south end (red-line 3, R0 kept); cleanouts, valves, balcony spouts and condensate drains (not modelled).
- **Electrician:** cable sizes, load sanction and testing.
- **Vastu consultant:** re-check of R1 before freeze.
- **Main site questions still open** in ASSUMPTIONS (top): A-15 foyer entry door, A-24 short parking bay, B-06 balcony merge.

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8. Design language and quality bar

- **Palette:**
  - light beige Italian marble floors in living rooms and bedrooms; warm-white walls
  - walnut and fluted walnut feature walls (foyer, lounge west wall, master headboard wall)
  - matte large-format porcelain in washrooms; brass accents; black metal frames
  - warm 2700 K-look cove lighting at dusk
- **Furniture:** Indian sizes (queen bed 1524 × 1981 mattress, king 1829 × 1981). All furniture is parametric with bevelled boxes, so keep new pieces in the same style: clean, slightly rounded, with walnut, marble, brass and linen.
- **Reference-driven pieces:** when the user shares a reference photo (the mandir, the tulsi), model its main features faithfully:
  - **Mandir:** marble back panel; backlit cusped (Mughal) arch; gold lotus reliefs; deity on a ledge; floating fluted cabinet with a marble top; brass diya stands; LED coves; pouf and rug in front.
  - **Tulsi:** floating marble plinth with LED under it; carved pedestal and planter; brass diyas; bamboo planters on each side; fluted backdrop with a brass Om.

  - **Verify both by eye:** render day and dusk in walk view, and iterate until it reads like the photo (proportions, glow, motifs).
- **Verification standard used so far:** after every change,
  1. regenerate, and run `check_dims` (main and R1) and `check_services`
  2. render the SVG plans to PNG and look at them
  3. render the 3D views headless (overview, the affected room in walk mode, day and dusk)
  4. on phones, check portrait at 390 × 844 with touch
  5. confirm the main site is unchanged
- **Report to the user:**
  - Briefly, with screenshots sent as files.
  - In plain language: what changed, where it is, any deviations with the reason, and any open questions.
  - Use feet-inches with metres for R1.

---

9. Environment notes and traps (learned the hard way)

- **Headless rendering:** use Playwright's Chromium with SwiftShader WebGL:

```
        import { chromium } from '/opt/node22/lib/node_modules/playwright/index.mjs';   // global
install in the cloud sandbox
        const b = await chromium.launch({ args: ['--use-gl=angle', '--use-angle=swiftshader', '--enable-
unsafe-swiftshader'] });
        // start `npx vite preview --port 4187 --strictPort` in gh18-house/viewer, then:
        await page.goto('http://localhost:4187/v1/?headless');
        await page.waitForFunction(() => window.viewer && window.viewer.ready, null, { timeout: 120000
});
        await page.evaluate(() => { viewer.setServices('electrical'); viewer.cine({ pos: [-12.8, 17,
33.8], target: [4.7, 3.2, 12.8] }); viewer.settle(2); });
        await page.evaluate(() => { viewer.walkPose(5400, 8350, 0.45, 90, -2); viewer.settle(3); });
// e.g. the mandir, looking east
        await page.screenshot({ path: 'out.png' });
```

  - The project pins `playwright 1.48.2` in devDependencies; in the cloud sandbox that version could not find a browser.
  - **Workaround used:** temporarily move `viewer/node_modules/playwright` aside while running scripts, so the global install is used, then **put it back**.
  - `scripts/shots.mjs` falls back to the global path automatically.
- **Render SVG plans to PNG:** load the SVG in Chromium with `page.setContent` and screenshot the `svg` element.
- **Network policy in the original sandbox:** blocked polyhaven.com, ambientcg, jsDelivr/unpkg CDNs, sketchfab.com and the deployed github.io site. That is why:
  - the textures are procedural
  - the Baleno was uploaded by the user as a zip
  - deploys can't be checked from the sandbox (use the GitHub Actions run list instead)
- **Don't use `pkill -f` with broad patterns.** It killed the agent's own shell once. Kill by PID.
- **Netlify vs GitHub Pages:** both build from the repo's `main` branch. GitHub Pages deploys on each push to `main` (Actions). If the user sees an older version on Netlify, ask them to check Netlify → Deploys, use "Clear cache and deploy site" if needed, and hard-refresh.
- **Deploying from a new branch:**

  - **Path filter:** the workflow only runs when files under `gh18-house/**` (or the workflow itself) change. A branch pushed with no new changes (e.g. a freshly created `main` identical to the default) triggers nothing. Start it by hand with the workflow's *Run workflow* button (`workflow_dispatch`).
  - **Environment rule:** the `github-pages` environment has a deployment-branch rule. When `main` was created, its deploy job was rejected within a second, with no log, until the owner allowed `main` under Settings → Environments → `github-pages` (and made `main` the default branch). If a deploy job fails instantly with no log, check that rule first.
- **Branch housekeeping from the agent sandbox:** pushing new branches works. Deleting a remote branch was refused (the connection dropped twice), so ask the user to delete branches in the GitHub UI (Branches page → bin icon).
- **The previous agent's render scripts** (`r1_walk.mjs`, `el_shot.mjs`, `svc_ui.mjs`, …) lived in a session scratchpad and are NOT in the repo. Recreate them from the snippet above, or add a `scripts/render.mjs` to the repo.
- **Client privacy:**
  - The client's name and address are in the JSON `meta` and are stripped from public builds (PUBLIC_BUILD=1).
  - Don't publish personal data elsewhere.
- **Licences:** Baleno CC BY 4.0 (credit must stay); procedural textures CC0; see `assets/LICENSES.md`.

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10. How the user likes to work

- **Who:** the user is the house owner's side, non-technical but precise about the home. They often write from a phone, sometimes in Hinglish (e.g. "ki jagah" = "instead of", "chotu sa" = "small").
- **What they value:**
  - quality ("No compromise on quality")
  - the old version kept intact (new work on new URLs)
  - Vastu-correct placements
  - feet-inches dimensions
  - seeing the result (screenshots)
- **Process they asked for:**
  - brainstorm and propose first when the change is big
  - then "go ahead" / "continue", step by step
  - they confirm interpretations, so ask when something is ambiguous
- **Running things themselves:** they offered to run things locally if that saves time and tokens. Suggest local commands when useful, e.g. `npm run dev` to preview.
- **Reports:** short; what changed, where, why, and any deviation or open question. Send renders. Don't bury the important caveat.

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11. Glossary

- **Areas and rooms:**
  - **OTS:** open-to-sky shaft (east side, x < 1031, y 10751–12665); carries stacks, risers and vents.
  - **Brahmasthan:** the Vastu centre of the house; kept free of DBs and heavy items.
  - **Mandir:** home shrine. **Tulsi vrindavan:** a raised stand for the holy basil plant. **Sethi / settee:** low seat with bolsters. **Diya:** oil lamp.
  - **Vastu zones:** N / NE / E / SE / S / SW / W / NW / centre on a 3 × 3 grid.

- **Levels:** FFL finished floor level, NGL natural ground level, POP plaster-of-Paris false ceiling.
- **Plumbing:**
  - UGT / OHT: underground / overhead water tank
  - RWH: rainwater harvesting pit; RWP: rainwater pipe
  - IC: inspection chamber; CO: sealed cleanout; GT: grease trap
  - SS: soil stack; DT: down-take (cold supply from the OHT); R-1: pumped riser
- **Electrical:**
  - DB: distribution board; MCB: miniature circuit breaker; RCD: 30 mA residual-current device
  - ODU: AC outdoor unit; TR: AC tonnage; EF: exhaust fan; VDP: video door phone
  - SB: switchboard; 2W: 2-way switch; IP65: weatherproof
- **Parties:**
  - RWA: residents' welfare association; MCG: Municipal Corporation of Gurugram
  - GH-17 / GH-19: the neighbouring houses on the east and west party walls
- **Item tags:**
  - A-xx / B-xx / C-xx / D-xx: main-site assumption and change IDs (ASSUMPTIONS A–D)
  - A1…E14, B15: R1 items (ASSUMPTIONS E)
  - red-line 1–18, T: Vastu register items
  - P1–P10: plumbing clouds; A1–A8 and B1–B9: electrical clouds (these electrical "A/B" tags are not ASSUMPTIONS IDs; the context makes clear which is meant)

Appendix A — Vastu red-line register (consultant, overlaid on R0; N = road side)

Type: R = retain, M = modify, H = hold point.

#	Space / zone	Type	Instruction	Structural / plumbing hold point
1	GF entrance foyer, N	R	Retain the N entry. Light-coloured main door. Foyer bright and uncluttered (N in deficit on GF).	—
2	GF guest room, NE / ENE	R	No heavy wardrobe on the NE wall; wardrobe on the S or W wall. Puja niche preferred in the NE corner.	—
3	GF toilet, E / ESE	M	WC at the south (ESE) end; wash / vanity at the north end.	Soil stack aligned with the FF toilet above (item 14); falls to a rear drain with inspection chambers.
4	GF electrical panel, centre	M	Move the DB / panel out of the Brahmasthan, preferably to the SE (kitchen side). Keep the centre light and clear.	Electrical consultant to confirm cable routing and panel access.
5	GF rear lawn, S / SW	M	Tall trees and heavy planters along the S and SW boundary. No water body, bore well or soak pit in the rear lawn.	Tree roots min. 1.5 m clear of footings and the rear drain.
6	GF front lawn, NE	M	RWH pit / water feature in the NE front lawn.	Min. 1.5 m clear of footings; waterproofing and overflow to the storm drain.
7	GF kitchen, SE	R	Retain. Hob on the SE wall (cook facing E); sink on the N side of the kitchen. Grease trap before the rear drain.	FF master toilet sits above the kitchen (see item 11).
8	GF study, NW / NNW	R	Retain. Desk facing N or E. Light neutral colours; no heavy storage (NNW in excess).	—

#	Space / zone	Type	Instruction	Structural / plumbing hold point
9	GF lift, W	H	Retain the location.	Lift vendor: pit depth, shaft and overrun; pit waterproofing; clash check with footings and drains.
10	GF OTS, E	H	Retain.	Structural design of the slab opening and trimmer beams before demolition.
11	FF master toilet, SE / SSE	M	Revised from the earlier toilet–dresser swap (the dresser zone is too small for a toilet). Put the WC at the ESE (dresser) end of the existing toilet, shower and vanity at the SSE end; the toilet stays within the existing sunken area.	No slab cutting needed. The toilet is above the GF kitchen: full waterproofing with a flood test; no soil pipes running exposed through the kitchen ceiling.
12	FF dresser, ESE	R	Retain as drawn. Mirror on the N or E wall.	—
13	FF master bedroom, SW / S	R	Headboard on the S wall; wardrobes on the S / SW walls (SSW in excess suits heavy storage).	Stone flooring load check on the FF slab.
14	FF toilet (centre-east), centre / E	R	Retain: it sits directly over the GF toilet, allowing a single vertical stack.	Shared soil / waste stack with the GF toilet; vent to the terrace.
15	FF toilet (bedroom 3), WNW	R	Retain: an ideal toilet zone.	Separate stack on the W side.
16	FF front curved balcony, N / NW	H	Keep open; light planters only.	Cantilever check; glass railing fixing and load; drainage slope.
17	FF stair / lift, W	R	Retain.	Headroom and lift overrun at the terrace.
18	FF rear balcony, SW	H	Retain.	Waterproofing, slope 1:100, overflow spout; railing fixing.
T	Terrace (not on R0), SW / W	H	Overhead water tank on the SW / W; open terrace kept on the N / NE. Any new room only in the S / SW.	Tank on an RCC stand over column / beam lines; check residual pressure at the FF showers.

Issue status: NOT FOR CONSTRUCTION. This is an owner instruction to the architect. An R1 construction issue is to be prepared by the architect, with structural and plumbing drawings. The Vastu consultant re-checks R1 before freeze.

Appendix B — Plumbing Layout Rev A (sheets 2–3, condensed)

Terrace plan (sheet 2):

- Heavy items go on the S / SW: OHT, solar heater, ODUs. N / NE are kept open and light.
- **OHT:** 1500–2000 L in the SW corner, on an RCC stand spanning column and beam lines. A float switch controls the pump PP; overflow to an RWP. The base must be at least 2.5–3 m above the FF shower heads, or there is a booster pump on DT-1.
- **DT-1 / DT-2:** both start at the OHT with a gate valve and a non-return valve.
  - DT-1 runs down the OTS: GF toilet, kitchen, utility, FF centre and master toilets.
  - DT-2 runs to the SS-3 boxed duct (FF WNW toilet).

- **Vents:** SS-1, SS-2 and SS-3 extend at least 1 m above the terrace finish, with a cowl, at least 3 m from any window or the solar heater.
- **Rain:** terrace slope 1:100 to the N; four 100 mm RWPs on the front face to the RWH pit; parapet spouts as overflow.
- **Waterproofing:** membrane plus brickbat coba; 48 h flood test; 300 mm upstands.

Stacks (sheet 3, A):

- **SS-1:** OTS, east side. Serves the GF toilet and the FF centre toilet. 110 mm soil, 75 waste, 50 vent. Discharges to IC-2 in the OTS.
- **SS-2:** rear face, just inside the E corner. Serves the FF master toilet. Runs in the sunken floor, not through the kitchen ceiling. Discharges to IC-1 in the rear lawn.
- **SS-3:** boxed duct inside the WNW toilet / study NW corner. Serves the FF WNW toilet. Discharges to IC-4 in the parking.
- **Kitchen:** sink + dishwasher, 50 mm waste, to a grease trap in a rear-lawn chamber, then IC-1.
- **Utility:** rear lawn (covered), to IC-1.
- **Rain:** four RWPs on the front face to the RWH and recharge pit in the front lawn; overflow to the storm drain.

Routing (B):

- Sewer: IC-1 → under the kitchen's E edge (sleeved) → IC-2 (OTS) → under the GF toilet and lobby / foyer (sleeved, sealed access CO-1 / CO-2) → IC-3 (front court) → IC-5 (parking).
- SS-3 → IC-4 → IC-5. Nothing under the guest room.
- Gradients: 150 mm uPVC SN8 at least 1:60; 110 mm at least 1:40.
- Chambers only in lawns, the OTS, the courtyard and the parking; heavy-duty covers at IC-4 / IC-5.
- Only IC-5 connects to the municipal sewer; no sewage to the storm drain.
- Everything stays inside the plot; nothing on the E / W party walls.

Water (C):

- **UGT:** 5000 L in the NE front lawn, at least 2 m from any sewer line or chamber and from the RWH pit.
- **Inlet:** from the gate with a meter and check valve, straight to the UGT.
- **Pumping:** PP 0.5–1 HP; R-1 (32 mm) in the E-wall chase and the OTS to the OHT.
- **Valves:** each toilet has an isolation valve.
- **Hot water:** a geyser per toilet, inside the toilet, high, out of the splash zone; optional solar heater on the S of the terrace.

Change log (D):

- **P1:** SS-2, IC-1 and the grease trap moved inside the plot; RCP-1 removed.
- **P2:** kitchen waste to the grease trap.
- **P3:** sewer re-routed under the toilet, lobby and foyer; R-1 into the E-wall chase.
- **P4:** UGT in the NE with ≥ 2 m clearances; IC-3 moved to the front court; the inlet doesn't cross the sewer.
- **P5:** no sewer branch into the storm drain; only IC-5; heavy-duty covers.
- **P6:** no pit or water body in the rear lawn; terrace rain to the front.
- **P7:** master toilet geyser inside, cleanout on the WC run, access panel at the bath, DT-1 branch in a wall chase with a valve.
- **P8:** SS-3 in a boxed duct.
- **P9 / P10:** front balcony RWP to the RWH; rear balcony spout to the lawn.

Assumptions: the E and W walls are treated as party walls; the municipal connection is at the front road; the OTS is large enough for SS-1, DT-1, R-1 and IC-2.

Appendix C — Electrical Layout Rev A (sheet 2, condensed)

Point schedule per room (A):

Room	Lights	Fan	6 A	16 A	AC	Geyser	EF	TV / data	Remarks
GF drawing / dining	4 + pendant	1	2	—	1 (2 TR)	—	—	TV + data	TV & AC on the W wall; 2-way switches at both ends; smoke detector
GF kitchen	2 + under-cabinet	—	1 + chimney	3	—	—	1	—	Hob on the E wall; EF on the rear wall; MW, DW, fridge on 16 A; heat detector
GF toilet	2	—	1	—	—	1	1	—	EF on the OTS wall; socket ≥ 600 mm from the basin (RCD); switchboard outside
GF lobby / stair / panel room	2	—	—	—	—	—	—	rack	2-way stair switching; inverter + battery + network rack; smoke detector
GF guest room	2	1	3	—	1 (1.5 TR)	—	—	TV	AC on the N wall, ODU on the front lawn; TV on the W wall; 2-way both sides
GF foyer	2	—	—	—	—	—	—	—	VDP screen; outdoor VDP at the gate
GF study	2	1	2 + UPS	—	1 (1.5 TR)	—	—	data	Desk on the E wall; sockets, data and UPS at the desk
External / parking	8 garden + 2 gate	—	1 IP65	—	—	—	—	2 CCTV	Rear CCTV; irrigation pump point; EV 32 A; meter at the gate
FF master bedroom	3	1	3 + 2 USB	—	1 (2 TR)	—	—	TV	Headboard on the W wall; 2-way both sides; AC moved to the W wall
FF master toilet	2	—	1	—	—	1	1	—	Switchboard outside; geyser high on the rear wall
FF dresser	1 + mirror	—	1	—	—	—	—	—	DB-2 on the E wall, stacked over DB-1
FF centre toilet	1 + mirror	—	—	—	—	1	1	—	EF on the OTS wall; switchboard outside in the lobby
FF lobby / stair	2	—	—	—	—	—	—	—	2-way stair switching; smoke detector
FF bedroom 2	2	1	3	—	1 (1.5 TR)	—	—	TV + data	Confirm the ODU route (E wall may be a party wall)
FF toilet (WNW)	1	—	—	—	—	1	1	—	EF on the external W wall; geyser on the inner wall
FF bedroom 3	2	1	3	—	1 (1.5 TR)	—	—	TV	Headboard on the W wall; 2-way both sides
FF balconies	3 + 1 wall	—	2 IP65	—	—	—	—	—	Weatherproof fittings
Terrace	2	—	1 IP65	—	—	—	—	—	ODU zone S / SW; pressure pump; solar inverter; spare conduit to the main panel

Circuits (B):

Circuit	Use	MCB	Cable (Cu, FRLS)
L1–L4	Lighting + fans, ≤ ~10 points each	6 / 10 A	1.5 sq mm
S1–S4	6 A sockets by area, 30 mA RCD	16 A	2.5 sq mm
P1–P3	Kitchen 16 A appliances	20 A	4 sq mm
G1–G4	Geysers	20 A	4 sq mm
AC1–AC6	ACs	20 A (2 TR: 25 A)	4 sq mm
EV / LIFT (3-phase) / PUMP	As per maker	As per maker	As per maker
UPS	Lights, fans, router, CCTV, study	16 A	2.5 sq mm

Load (C):

- lighting + fans 1.8 kW
- sockets 3.0 kW
- kitchen 4.5 kW
- geysers 4 × 2.0 = 8.0 kW
- ACs 10.5 kW
- EV 7.4 kW
- lift 4.0 kW
- pumps 1.5 kW
- **Total:** about 41 kW connected; 20–25 kW maximum demand, so 3-phase.

Clouds (D):

- **A1:** rear lawn IP65, irrigation pump point, rear CCTV, ODU for the drawing-room AC.
- **A2:** drawing room TV and AC on the W wall; pendant; 2-way at both ends; smoke detector.
- **A3:** kitchen hob and chimney on the E wall; EF to the rear wall; fridge and DW points; heat detector; DB-1 in the SE.
- **A4:** GF toilet EF to the OTS wall; socket away from the basin; switchboard outside.
- **A5:** DB-1 out of the centre; 2-way stair switching; inverter + rack in the panel room; smoke detector.
- **A6:** guest room TV on the W wall; AC on the N wall with a front-lawn ODU; bedside points both sides.
- **A7:** study desk sockets, data and UPS on the E wall.
- **A8:** gate lights and outdoor VDP.
- **B1 / B9:** IP65 on the balconies.
- **B2:** master AC off the glazed wall; sockets / USB and 2-way both sides; smoke detector.
- **B3 / B4:** master toilet switchboard outside, geyser out of the splash zone; dresser mirror light; DB-2 over DB-1.
- **B5:** centre toilet EF on the OTS wall, mirror light, switchboard outside.
- **B6:** bedroom 2 ODU route to confirm.
- **B7:** WNW toilet EF and geyser swapped (EF on the external wall).
- **B8:** bedroom 3 sockets and 2-way both sides; smoke detector.

Typical heights: switchboard 1200, bedside 450, counter 1100, AC 2100, geyser 1800 mm above FFL.
Protection: 30 mA RCD on socket, wet, outdoor and EV circuits; SPD at the main panel; at least 2 earth pits plus a lift earth.

Note on room names: the sheet's FF "Bedroom 2" is R1's FF-BD3 (Bedroom 3, east side); the sheet's "Bedroom 3" is R1's kids room; the sheet's "Centre toilet" is R1's Bedroom 3 Toilet; the sheet's "Toilet (WNW)" is R1's kids toilet.

Appendix D — Key coordinates in R1 (mm)

GF rooms (bounding boxes, x0–x1 × y0–y1):

- Kitchen 230–3150 × 5665–11365
- Family Lounge 3300–9151 × 5665–8500
- Dining 3300–9151 × 8500–11365
- Lobby 4350–7200 × 10766–17196
- Toilet 230–4200 × 11515–13916
- Guest Room 230–4216 × 14066–19106
- Foyer 3229–5902 × 17375–20290
- Guest Lounge / Study 6080–9151 × 15274–19716
- Under-stair 6080–9151 × 12865–15124
- Lift 7350–9151 × 10916–12715
- OTS −435–731 × 10751–12665

GF outdoor areas:

- back lawn 230–9151 × 200–2814
- deck 230–9151 × 2814–5515 (pergola posts at y 2964, x 2489 / 4659 / 6830 / 9001)
- front lawn 230–3237 × 20153–21975
- entrance court 230–5900 × 19257–21975
- parking 6080–9151 × 19896–22276

FF rooms:

- Master Bedroom 3143–9151 × 5665–10669
- Dresser 230–2993 × 5665–8605
- Master Washroom 230–2993 × 8755–10669
- Balcony 5349–9151 × 4575–6655
- Bedroom 3 Toilet 1031–4269 × 10849–12813
- Lobby 4419–7198 × 10849–15306
- Bedroom 3 230–4269 × 12964–20006
- Kids Room 4371–9151 × 15456–21217
- Kids Toilet 6048–9151 × 15305–16805

SF rooms:

- Lobby 4419–7198 × 10849–15306
- Servant Washroom 1031–2268 × 10849–12813
- Store 2418–4269 × 10849–12813
- Servant Room 230–4269 × 12963–15306
- Back Terrace + Utility 230–9151 × 5665–10669
- Front Terrace 230–9151 × 15305–21217

Key items:

- mandir GF-F25 (3525, 8350), rot 270
- tulsi GF-O09 (645, 1650), rot 270
- OHT SF-R01 (6300, 11600); solar SF-R02 (3000, 11500), both on the roof
- car GF-O07 (7615, 21945)
- DB-1 (250, 9451); DB-2 (2973, 5950); DB-3 (250, 13300); INV (8525, 12885); MTR (5000, 22290)
- SS-1 (450, 11100); SS-3 (8950, 15420); UGT (900, 21100); RWH (4500, 21400)

Appendix E — Checklist for any change

1. **Understand the request.** Map it to the frame (R1: north = +y, east = −x). If it is ambiguous or conflicts with Vastu or the consultants' rules, ask one clear question with options.
2. **Pick the right place:**
   - main site → `build_house.py` (only if asked)
   - R1 → a patch in `build_r1.py`
   - visuals → `furniture.js`, `plans.py` symbol, `build.js` light
   - services → `e12_plumbing` / `e13_electrical`
3. **Rebuild and check everything:**
   - `build_r1.py`
   - `check_dims` for main and R1 (both PASS)
   - `check_services` (no FAIL; explain any new WARN)
   - `boq.py` and `html2pdf.mjs` (R1), so the materials schedule and the Materials view follow the change
   - `plans.py` for main and R1; main must be byte-identical unless the change was meant for it
4. **Build the viewer** (`npm run build`), render the affected views (day and dusk, walk and overview, phone if it's UI), and look at them.
5. **Update the records:** ASSUMPTIONS (row with the reasoning), CHANGE_REGISTER (if R1), README (if the workflow changed).
6. **Commit** with a descriptive message on `house-design-v1`, and **push**. Once the user approves, **merge into `main`**; GitHub Pages redeploys from `main` automatically.
7. **Report to the user:** what changed, where (feet-inches), why, any deviation or open question, with screenshots.